

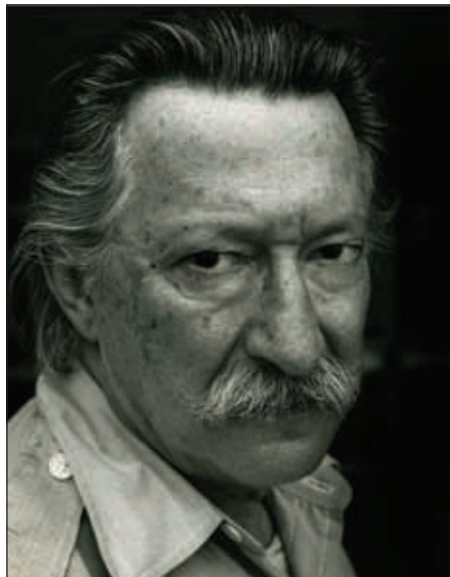
ELIZA, the first chatbot

In 1966, Joseph Weizenbaum created what is known to be the first chatbot. His purpose in creating the system was to pass the Turing Test successfully, for which he needed to “coach” his chatterbot to understand the English language in a natural form, along with being able to respond to it efficiently. He liked to compare this task to the protagonist of the film *My Fair Lady*, Eliza Doolittle – a Cockney girl who is taught by a linguist to speak properly. The work that Weizenbaum did ended up being the gold standard in the field of chatterbots and till today, modern chat bots are based on the same fundamental design.

To understand how ELIZA works, one needs a little bit of a background in understanding how humans grasp conversation. In his paper, *ELIZA: A Computer Program For the Study of Natural Language Communication Between Man and Machine* Weizenbaum notes that the psychology of human beings is such that we tend to give context to someone we’re conversing with. In other words, if a conversation with a machine were to appear human to us, we would want to personify it with human-like capabilities. This base psychological inclination of the human mind went a long way in making it easier for Weizenbaum to design ELIZA.

Of course, natural language isn’t an easy thing to understand, let alone program into a computer. To be successful, ELIZA would have to be considered a foreigner to the language. With this base assumption, Weizenbaum taught ELIZA a few keywords such as “you”, “I”, “me”, etc. This pre-fed keyword dictionary is the building block of how ELIZA works.

Much like a modern search engine, ELIZA identifies keywords and responds to them, uti-



Joseph Weizenbaum inventor of ELIZA chat bot